

HTML BASICS

UNIT 1

SOD316C / SSD316C

```
for mirror_object in mirror_mod.mirror_object:  
    operation == "MIRROR_X":  
        mirror_mod.use_x = True  
        mirror_mod.use_y = False  
        mirror_mod.use_z = False  
    operation == "MIRROR_Y":  
        mirror_mod.use_x = False  
        mirror_mod.use_y = True  
        mirror_mod.use_z = False  
    operation == "MIRROR_Z":  
        mirror_mod.use_x = False  
        mirror_mod.use_y = False  
        mirror_mod.use_z = True
```

```
selection at the end -add  
mirror_ob.select= 1  
modifier_ob.select=1  
context.scene.objects.active  
("Selected" + str(modifier_ob.name))  
mirror_ob.select = 0  
= bpy.context.selected_object  
data.objects[one.name].select  
print("please select exactly one mirror")
```

OPERATOR CLASSES -----

```
bpy.types.Operator):  
    X mirror to the selected  
    object.mirror_mirror_x"  
    mirror X"
```

```
def execute(self, context):  
    if context.object is not None:
```

HTML BASICS



HTML BASICS

- Welcome to HTML Basics
- This introductory unit leads you through the basics of Hyper Text Markup Language
- You will learn to use HTML to author an HTML page to display in a web browser



OBJECTIVES

- Use a text editor to author an HTML document
- Be able to use basic tags to denote paragraphs, emphasis or special type
- Create hyperlinks to other documents
- Create an email link
- Add images to your document
- Use a table for layout
- Apply colors to your HTML document



PREREQUISITES

- Notepad and Visual Studio Code (VS Code)
- **NOTE:** Avoid using is a word processor (like Microsoft Word) for authoring your HTML documents.

WHAT IS HTML?

WHAT IS AN HTML FILE?

- HTML stands for Hyper Text Markup Language
- An HTML file is a text file containing small markup tags
- The markup tags tell the **Web browser** how to display the page
- An HTML file must have an **.htm** or **.html** file extension



TRY IT

- Open your text editor or preferably VS Code and type the following text:

```
<html>
```

```
<head>
```

```
<title>My First Webpage</title>
```

```
</head>
```

```
<body>
```

```
This is my first homepage. <b>This text  
is bold</b>
```

```
</body>
```

```
</html>
```

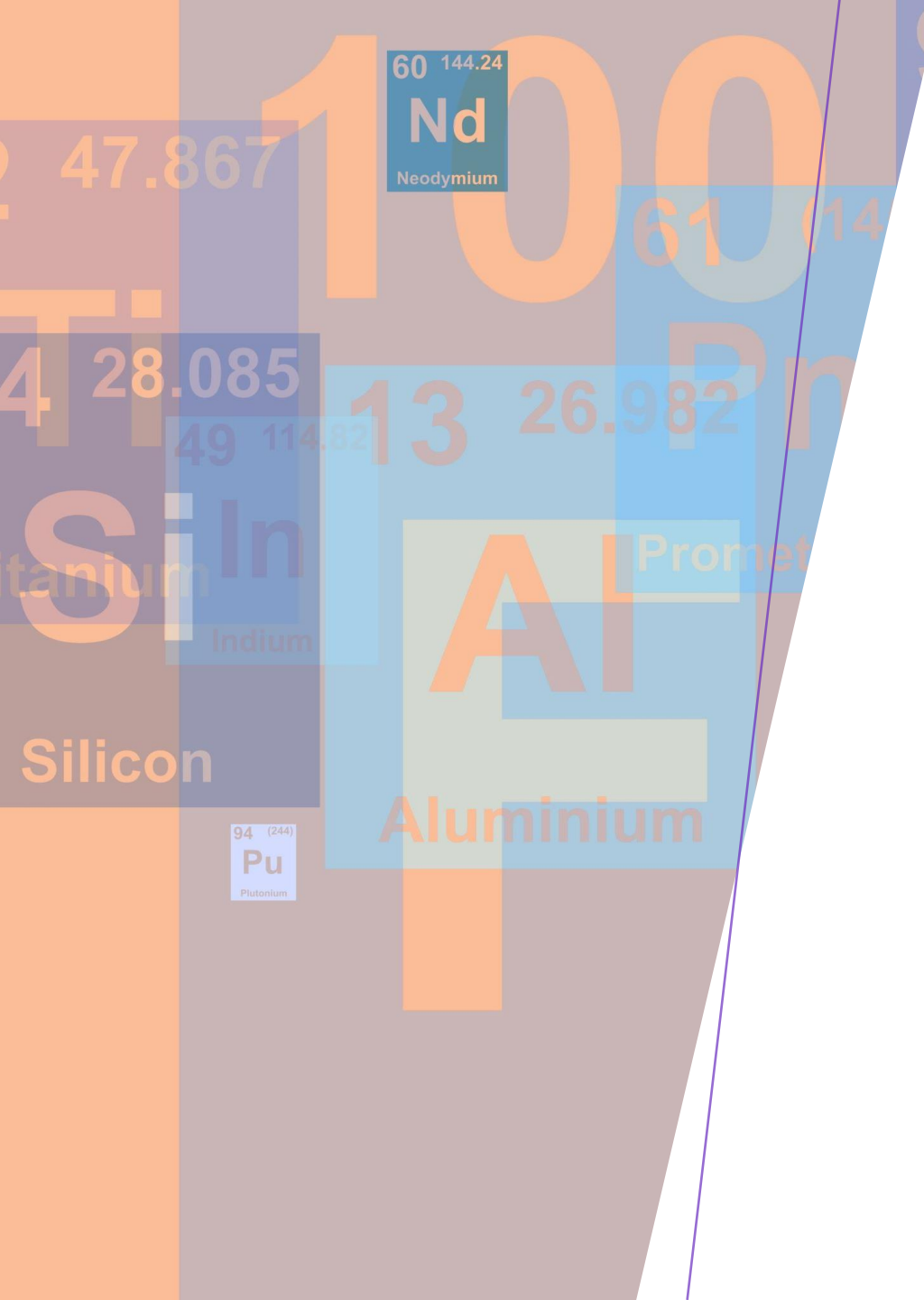
TRY IT

- Save the file as **mypage.html**
- Simply double-click on the file icon of the saved file or select Open in the File menu of your browser



EXAMPLE EXPLAINED

- This is the minimum required information for a web document and all web documents should contain these basic components
- The first tag in your html document is
- The `<title>` tag is used to uniquely identify each document and is also displayed in the title bar of the browser window



HTM OR HTML EXTENSION?

- When you save an HTML file, you can use either the **.htm** or the **.html** extension
- The **.htm** extension comes from the past when some of the commonly used software only allowed three letter extensions
- **mypage.htm** and **mypage.html** are treated as different files by the browser



HOW TO VIEW HTML SOURCE

- A good way to learn HTML is to look at how other people have coded their html pages
- To find out, simply click on the View option in your browsers toolbar and select Source or Page Source
- This will open a window that shows you the actual HTML of the page

HTML TAGS

HTML TAGS: WHAT ARE HTML TAGS?

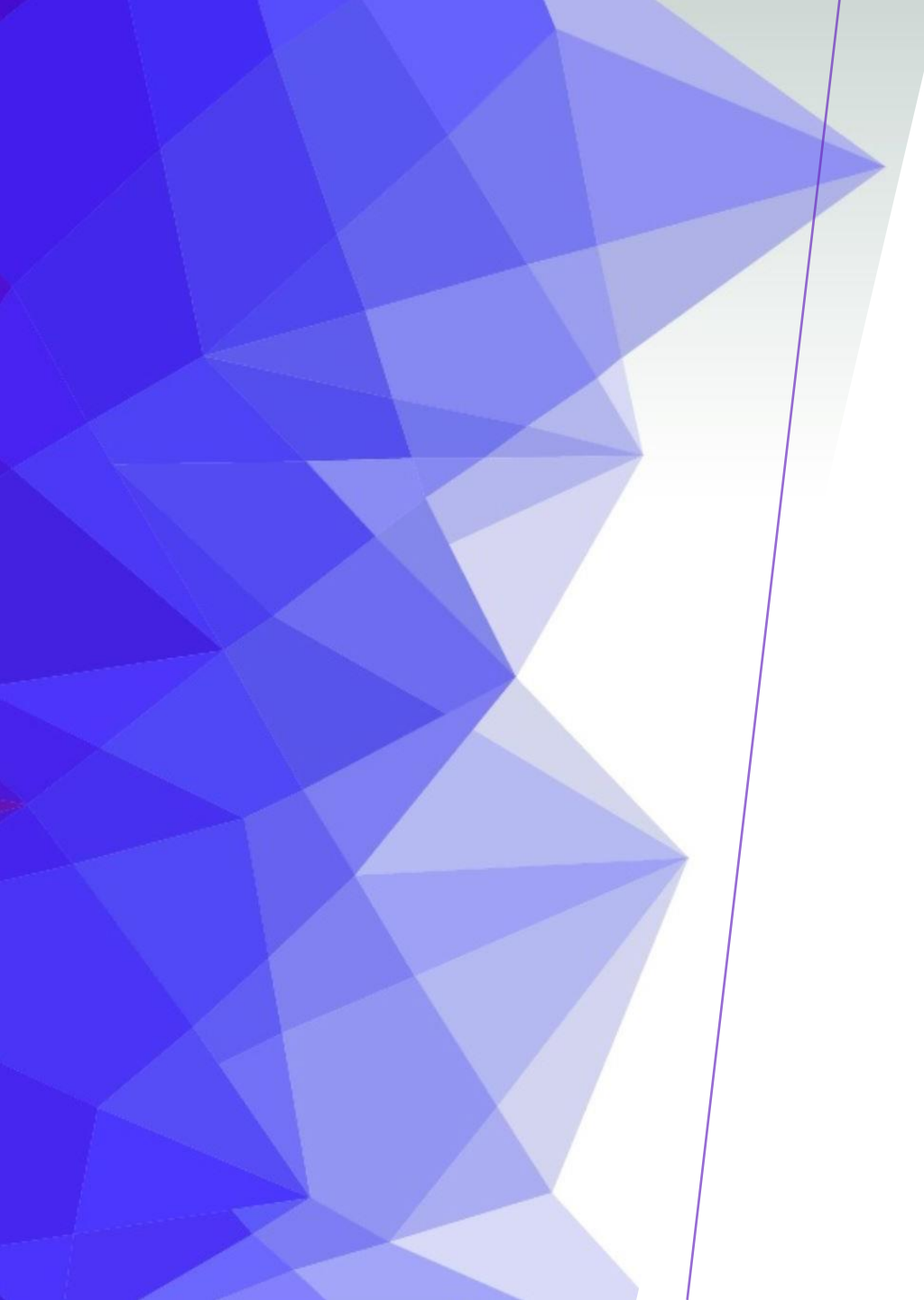
- HTML tags are used to mark-up HTML elements
- HTML tags are surrounded by the two characters `<` and `>`
- The surrounding characters are called angle brackets
- HTML tags normally come in pairs like `<b>` and `</b>`



HTML TAGS: WHAT ARE HTML TAGS?

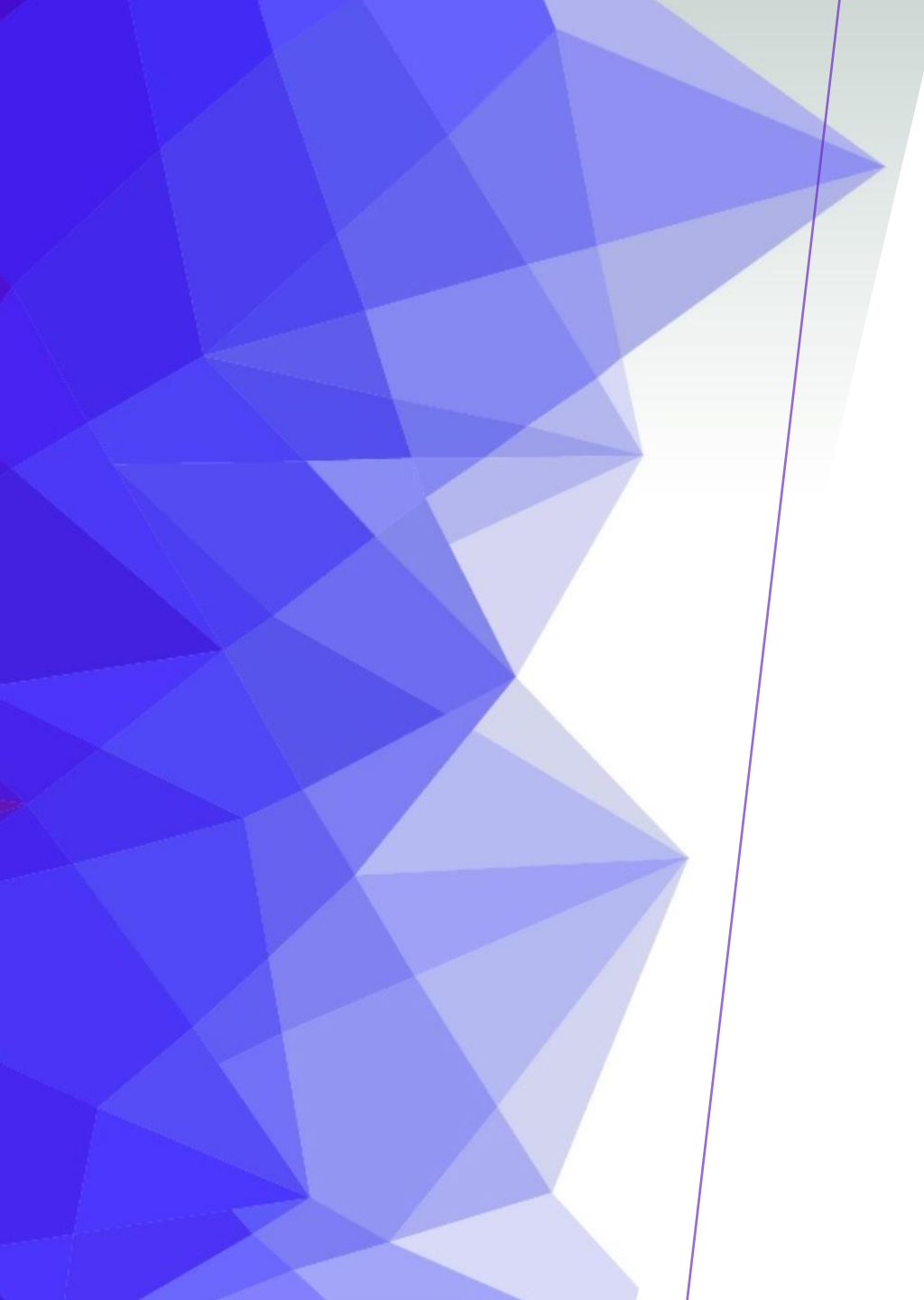
- The first tag in a pair is the start tag, the second tag is the end tag
- The text between the start and end tags is the element content
- HTML tags are not case sensitive, means the same as





LOGICAL VS. PHYSICAL TAGS

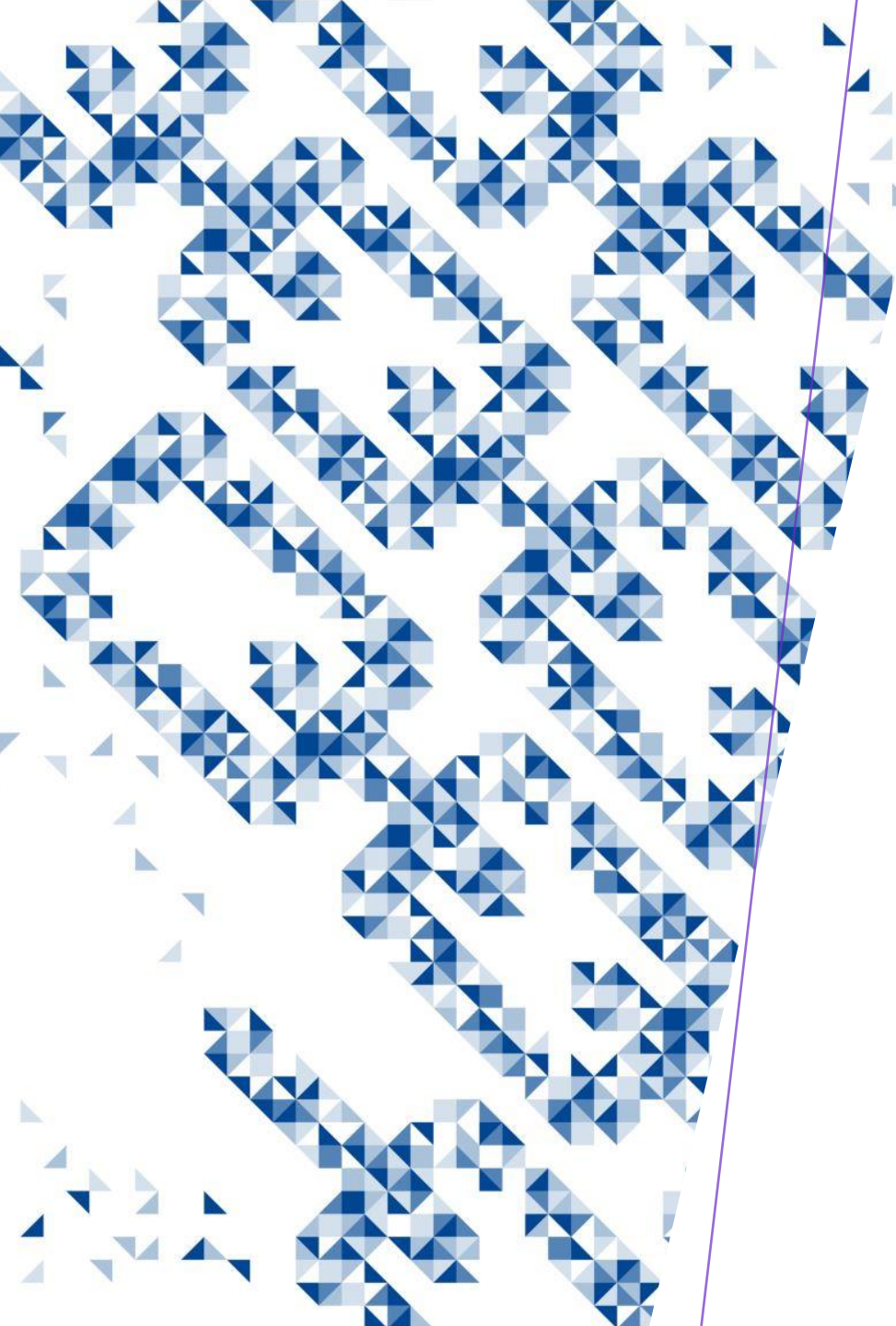
- In HTML there are both logical tags and physical tags.
- **Logical tags** are designed to describe (to the browser) the enclosed text's meaning.
- An example of a logical tag is the `<strong>` `</strong>` tag.
- By placing text in between these tags, you are telling the browser that the text has some greater importance.
- By default, all browsers make the text appear bold when in between the `<strong>` and `</strong>` tags (the `<b>` `</b>` tags will do the same).



LOGICAL VS. PHYSICAL TAGS

- **Physical tags** on the other hand provide specific instructions on how to display the text they enclose.
- Examples of physical tags include:
 - `<b>`: Makes the text bold.
 - `<big>`: Makes the text usually one size bigger than what's around it.
 - `<i>`: Makes text italic.
- Physical tags were invented to add style to HTML pages because style sheets were not around, though the original intention of HTML was to not have physical tags.
- Rather than use physical tags to style your HTML pages, you should use style sheets.

HTML ELEMENTS AND HTML TAGS



HTML ELEMENTS

- Remember the HTML example from the previous page?

```
<html>
```

```
<head>
```

```
<title>My First Webpage</title>
```

```
</head>
```

```
<body>
```

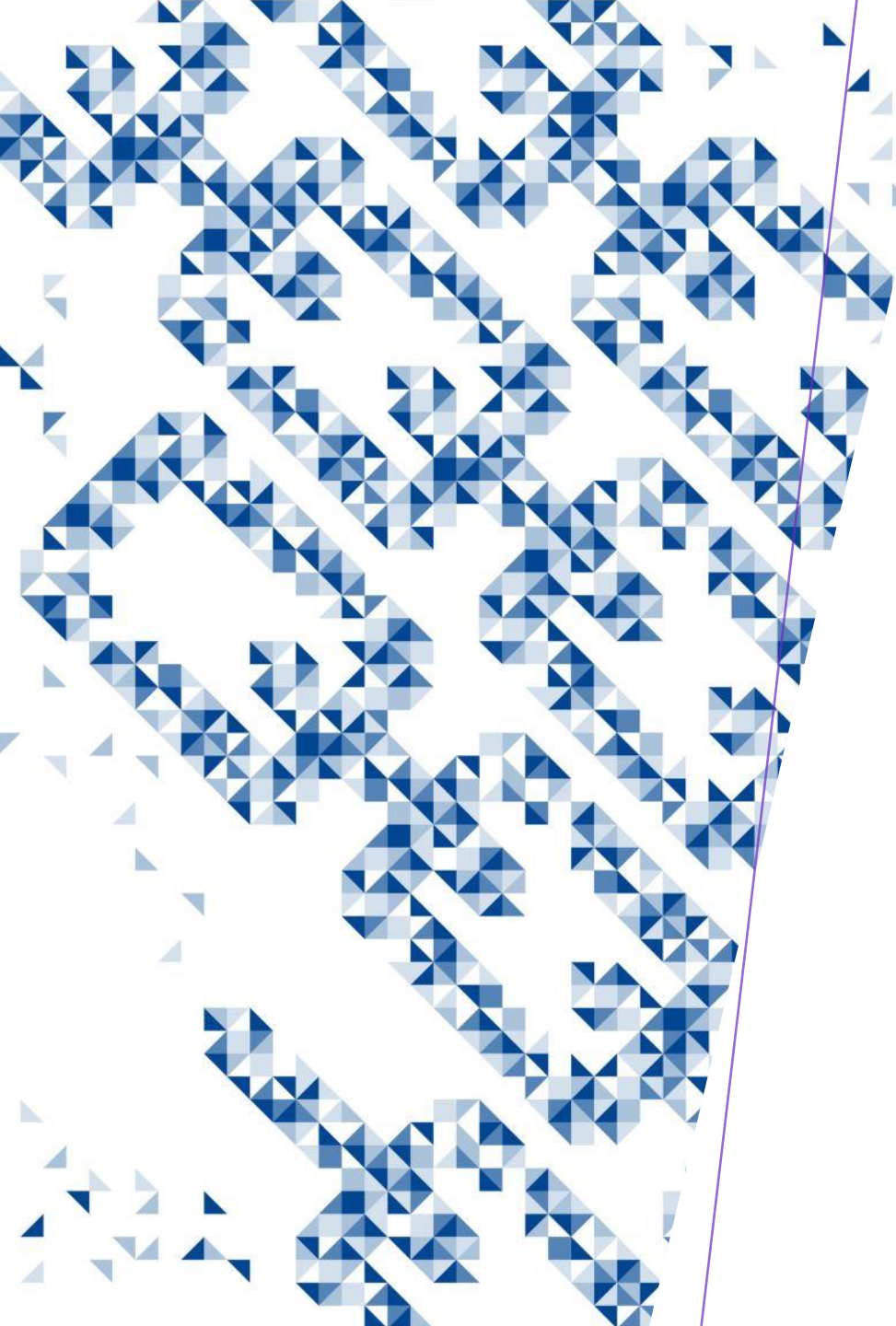
```
This is my first homepage. <b>This text is bold</b>
```

```
</body>
```

```
</html>
```

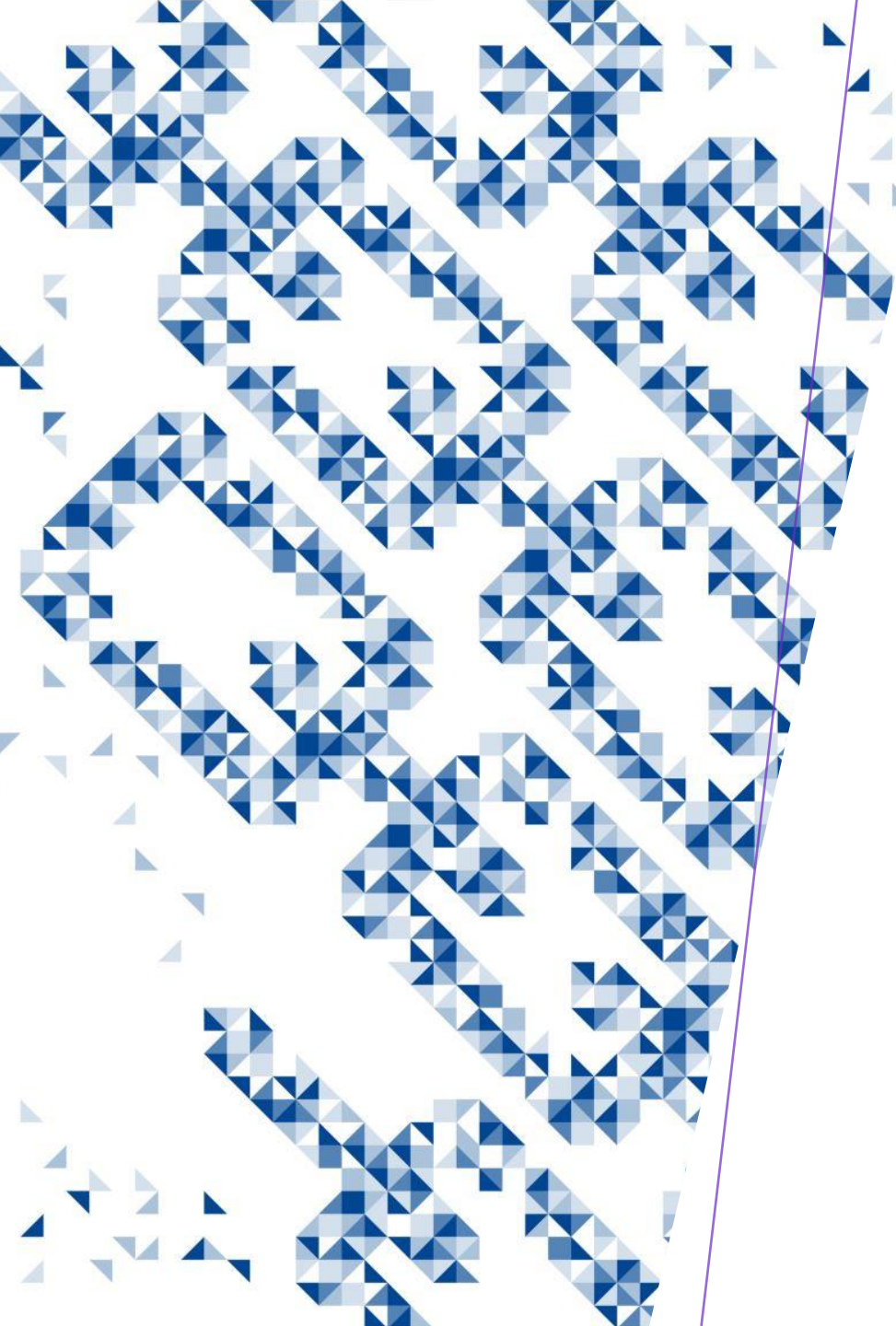
- This is an HTML element:

```
<b>This text is bold</b>
```



HTML ELEMENTS

- The HTML element begins with a start tag: `<b>`
- The content of the HTML element is: **This text is bold**
- The HTML element ends with an end tag: `</b>`
- The purpose of the `<b>` tag is to define an HTML element that should be displayed as bold.



HTML ELEMENTS

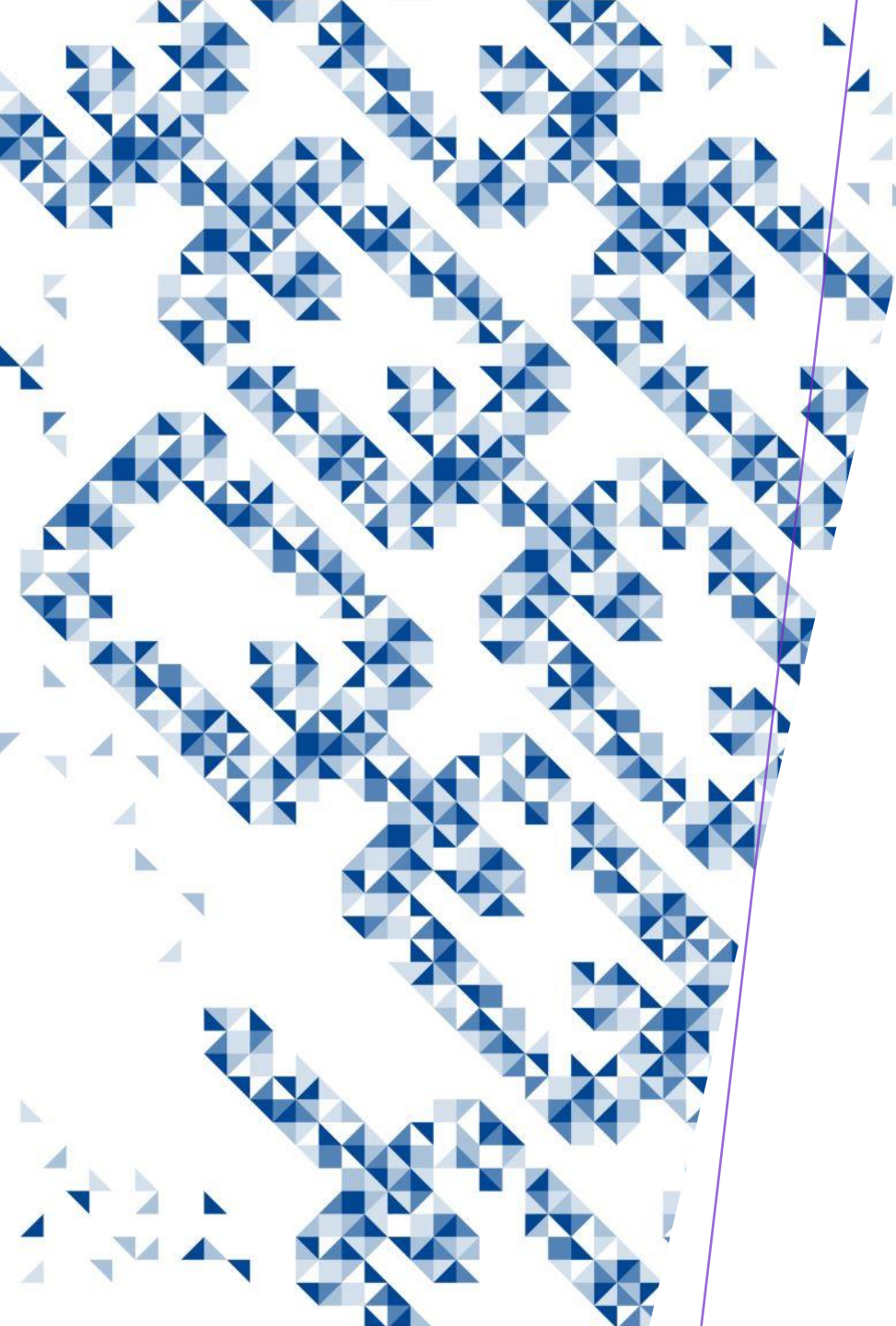
- This is also an HTML element:

`<body>`

`This is my first homepage. This
text is bold`

`</body>`

- This HTML element starts with the start tag `<body and ends with the end tag </body.`
- The purpose of the `<body tag is to define the HTML element that contains the body of the HTML document.`

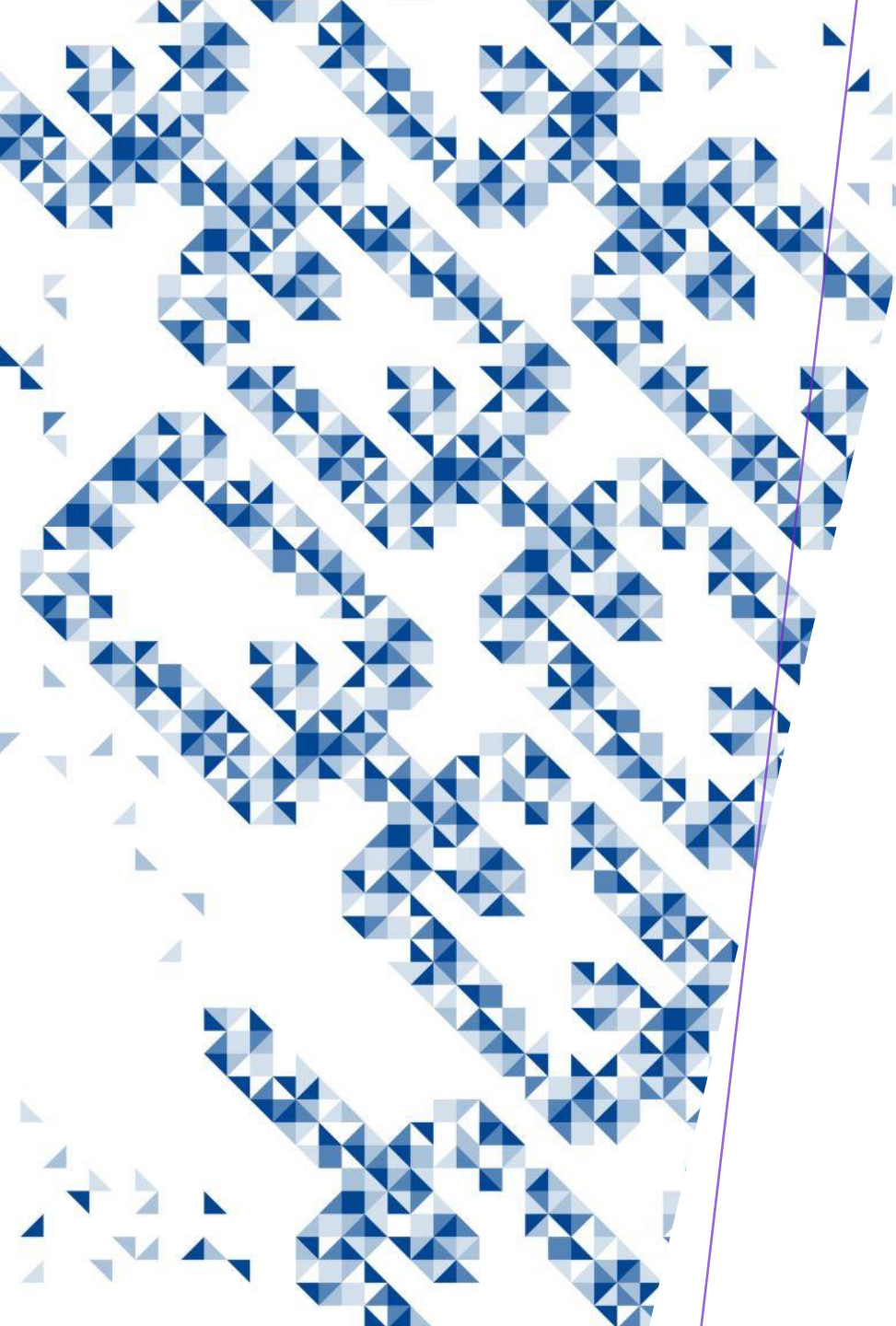


NESTED TAGS

- You may have noticed in the example above, the `<b>` tag also contains other tags, like the `<b>` tag.
- When you enclose an element in with multiple tags, the last tag opened should be the first tag closed. For example:

`<p><b><em>This is NOT the proper way to close nested tags.</p></em></b>`

`<p><b><em>This is the proper way to close nested tags.</em></b></p>`



NESTED TAGS

- **Note** that it doesn't matter which tag is first, but they must be closed in the proper order

A decorative graphic on the left side of the slide featuring a pile of 3D, light blue letters and symbols (including A, C, G, B, V, W, X, and numbers like 1, 2, 3, 4, 5, 6, 7, 8, 9, 0) scattered on a white surface. A thin purple line runs diagonally from the top right towards the bottom left, separating the graphic from the text area.

WHY USE LOWERCASE TAGS?

- You may notice we've used lowercase tags even though it was said that HTML tags are not case sensitive means, e.g., `<b>` the same as `<B>`.
- The **World Wide Web Consortium**, or **W3C**, the group responsible for developing web standards, recommends lowercase tags in their HTML 4 recommendation, and EXtensible HyperText Markup Language (XHTML) requires lowercase tags.

TAG ATTRIBUTES

- Tags can have **attributes**
- Attributes can provide additional information about the HTML elements on your page
- For instance, if we add the **bgcolor** attribute, we can tell the browser that the background color of your page should be blue, like this: **<body bgcolor="blue">**.



TAG ATTRIBUTES

- This tag, for instance, defines an HTML table:
<table>.
- With an added border attribute, you can tell the browser that the table should have no borders: **<table border="0">**.
- Attributes always come in name/value pairs like this: **name="value"**.



TAG ATTRIBUTES

- Attributes are always added to the start tag of an HTML element and the value is surrounded by quotes.
- Double style quotes are the most common, but single style quotes are also allowed.
- In some rare situations, like when the attribute value itself contains quotes, it is necessary to use single quotes:

**name= 'George "machine Gun"
Kelly'**



TAG ATTRIBUTES

- Browsers should continue to support deprecated tags and attributes, but eventually these tags are likely to become obsolete and so future support cannot be guaranteed.
- **Note** that some tags we will discuss are deprecated, meaning the World Wide Web Consortium (W3C) the governing body that sets HTML, XML, CSS, and other technical standards decided those tags and attributes are marked for deletion in future versions of HTML and XHTML.
- For a complete list of tags, visit W3C.org



BASIC HTML TAGS

- Headings, paragraphs and line breaks are some of the most commonly used and “important” HTML tags.

Tag	Description
<html>	Defines an HTML document
<body>	Defines the document's body
<h1> to <h6>	Defines header 1 to header 6
<p>	Defines a paragraph
 	Inserts a single line break
<hr>	Defines a horizontal rule
<!-->	Defines a comment

HTML HEADINGS



HTML TAGS: HEADINGS

- Headings are defined with the `<h1>` to `<h6>` tags. `<h1>` defines the largest heading while `<h6>` defines the smallest.

`<h1>This is a heading</h1>`

`<h2>This is a heading</h2>`

`<h3>This is a heading</h3>`

`<h4>This is a heading</h4>`

`<h5>This is a heading</h5>`

`<h6> This is a heading</h6>`



HTML TAGS: HEADINGS

- HTML automatically adds an extra blank line before and after a heading when viewed in a browser.
- A useful heading attribute is **align**:

```
<h5 align="left">I can align  
headings </h5>
```

```
<h5 align="center">This is a  
centered heading </h5>
```

```
<h5 align="right">This is a heading  
aligned to the right </h5>
```

HTML PARAGRAPHS



PARAGRAPHS

- Paragraphs are defined with the `<p>` tag. Think of a paragraph as a block of text.
- You can use the `align` attribute with a paragraph tag as well:

```
<p align="left">This is a  
paragraph</p>
```

```
<p align="center">this is another  
paragraph</p>
```



PARAGRAPHS

- **Important:** You must indicate paragraphs with `<p>` elements.
- A browser ignores any indentations or blank lines in the source text.
- Without `<p>` elements, the document becomes one large paragraph.
- HTML automatically adds an extra blank line before and after a paragraph.

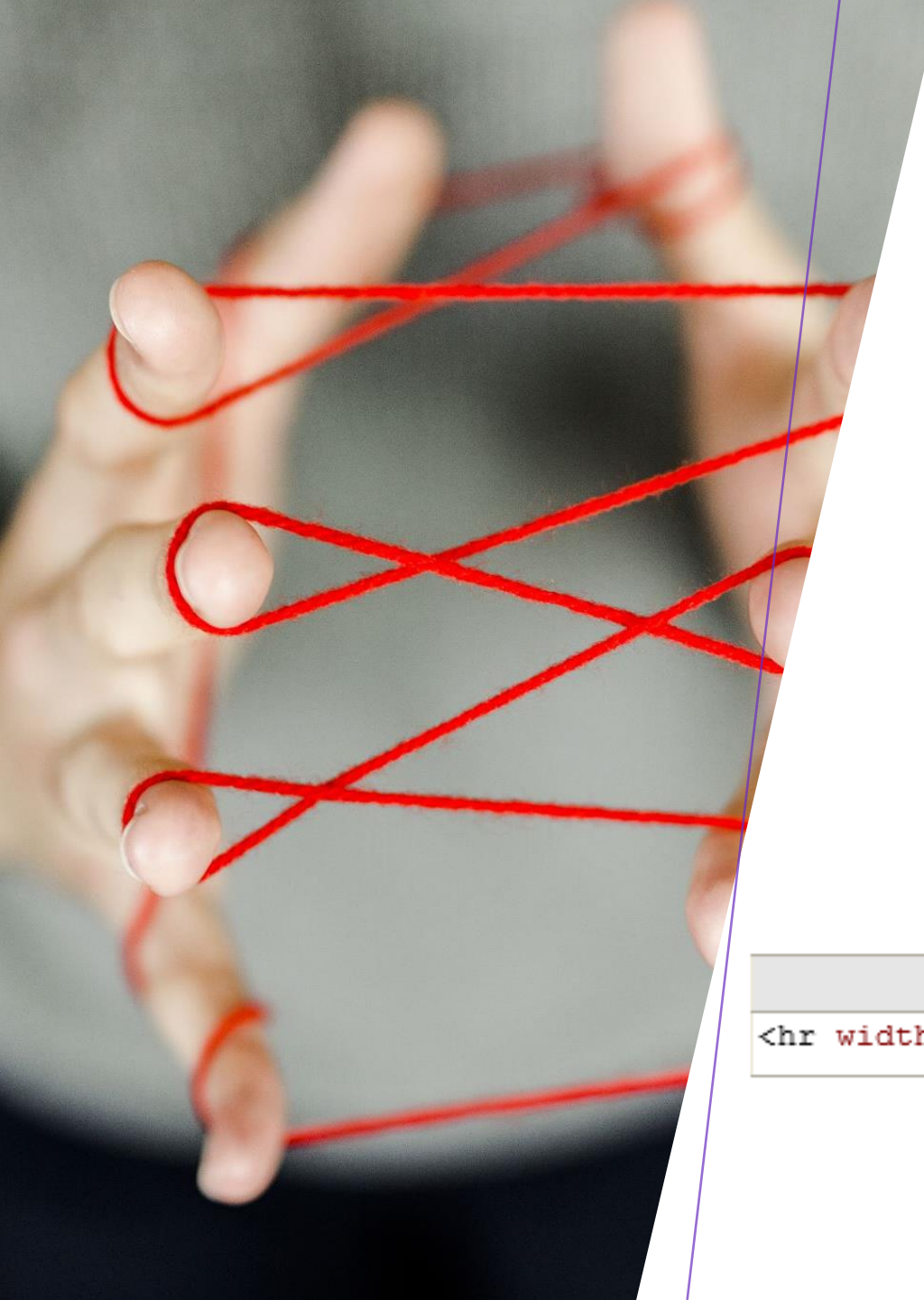
HTML LINE BREAKS

LINE BREAKS

- The **
** tag is used when you want to start a new line, but don't want to start a new paragraph.
- The **
** tag forces a line break wherever you place it. It is similar to single spacing in a document.
- **Note** that the **
** tag has no closing tag.

This Code	Would Display
<code><p>This
 is a para
 graph with line breaks</p></code>	This is a para graph with line breaks

HORIZONTAL RULE



HORIZONTAL RULE

- The **<hr>** element is used for horizontal rules that act as dividers between sections, like this:

- The horizontal rule does not have a closing tag.
- It takes attributes such as align and width. For instance:

This Code	Would Display
<code><hr width="50%" align="center"></code>	

COMMENTS

COMMENTS IN HTML

- The **comment tag** is used to insert a comment in the HTML source code.
- A comment can be placed anywhere in the document and the browser will ignore everything inside the brackets.
- You can use comments to write notes to yourself or write a helpful message to someone looking at your source code.

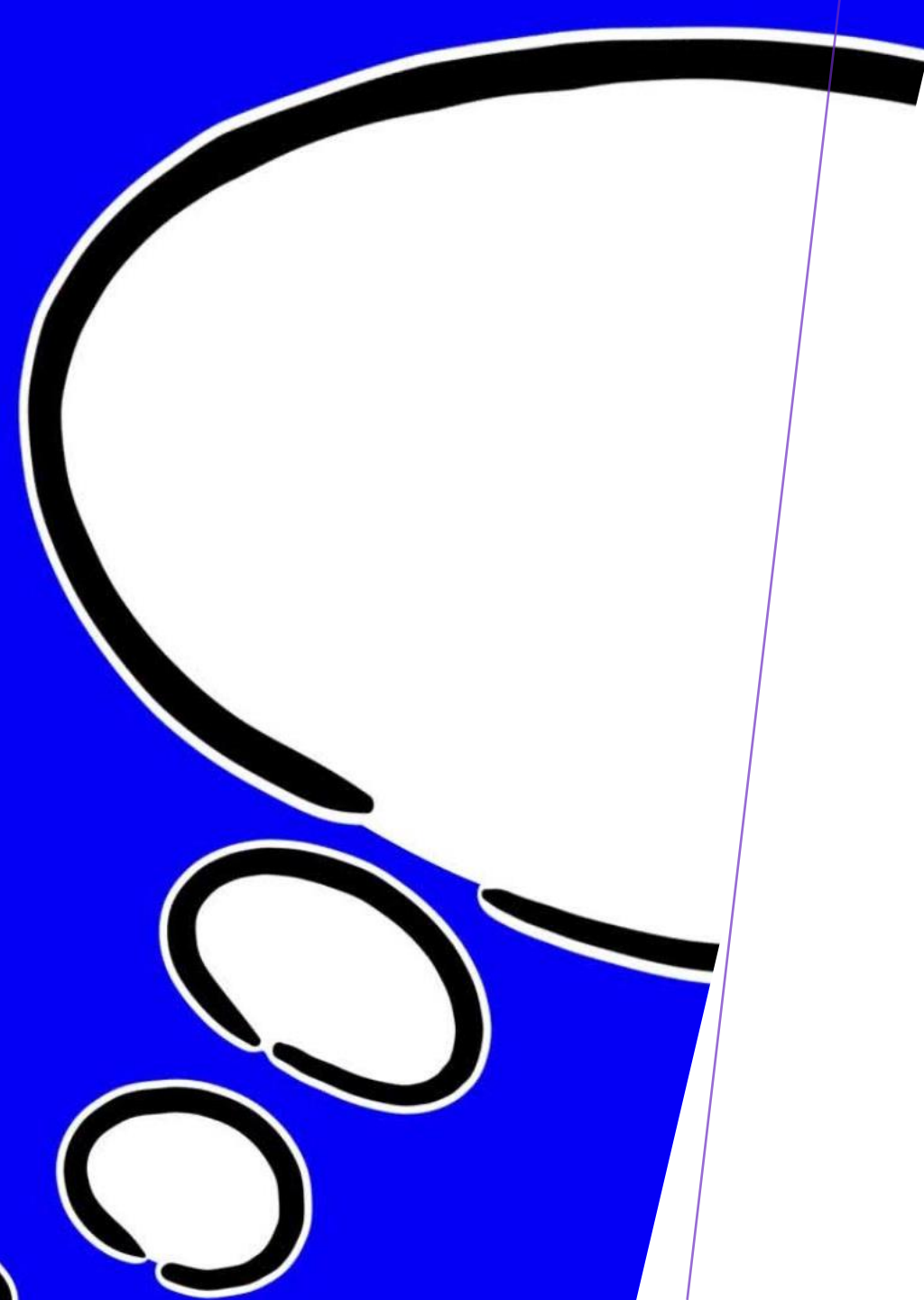
COMMENTS IN HTML

This Code	Would Display
<pre><p> This html comment would <!-- This is a comment --> be displayed like this.</p></pre>	This HTML comment would be displayed like this.

- **Note** that you don't see the text between the tags `<!--` and `-->`. If you look at the source code, you will see the comment.
- To view the source code for this page, in your browser window, select **View** and then select **Source**.
- You need an exclamation point after the opening bracket `<!--` but not before the closing bracket `-->`.

COMMENTS IN HTML

- HTML automatically adds an extra blank line before and after some elements, like before and after a paragraph, and before and after a heading.
- If you want to insert blank lines into your document, use the **
** tag.



TRY IT OUT!

- Open your text editor or IDE and type the following text:

```
<html>
```

```
<head>
```

```
<title>My First Webpage</title>
```

```
</head>
```

```
<body>
```

```
<h1 align="center">My First Webpage</h1>
```

```
<p>Welcome to my first web page. I am writing  
this page using a text editor and plain old  
html.</p>
```

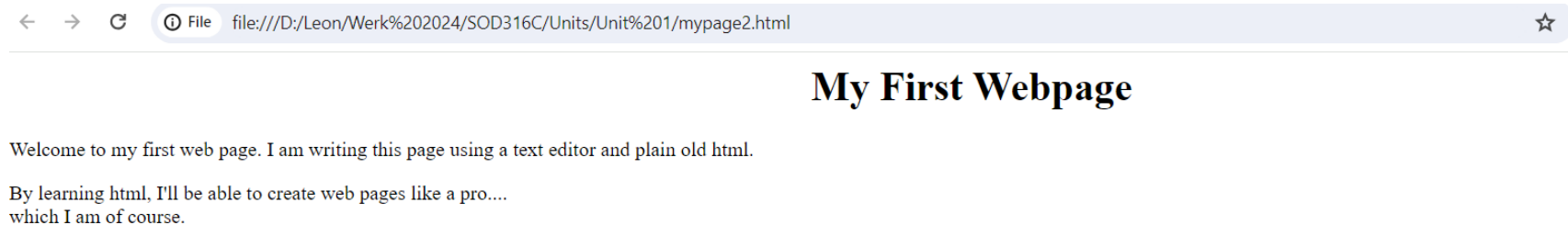
```
<p>By learning html, I'll be able to create  
web pages like a pro....<br> which I am of  
course.</p>
```

```
</body>
```

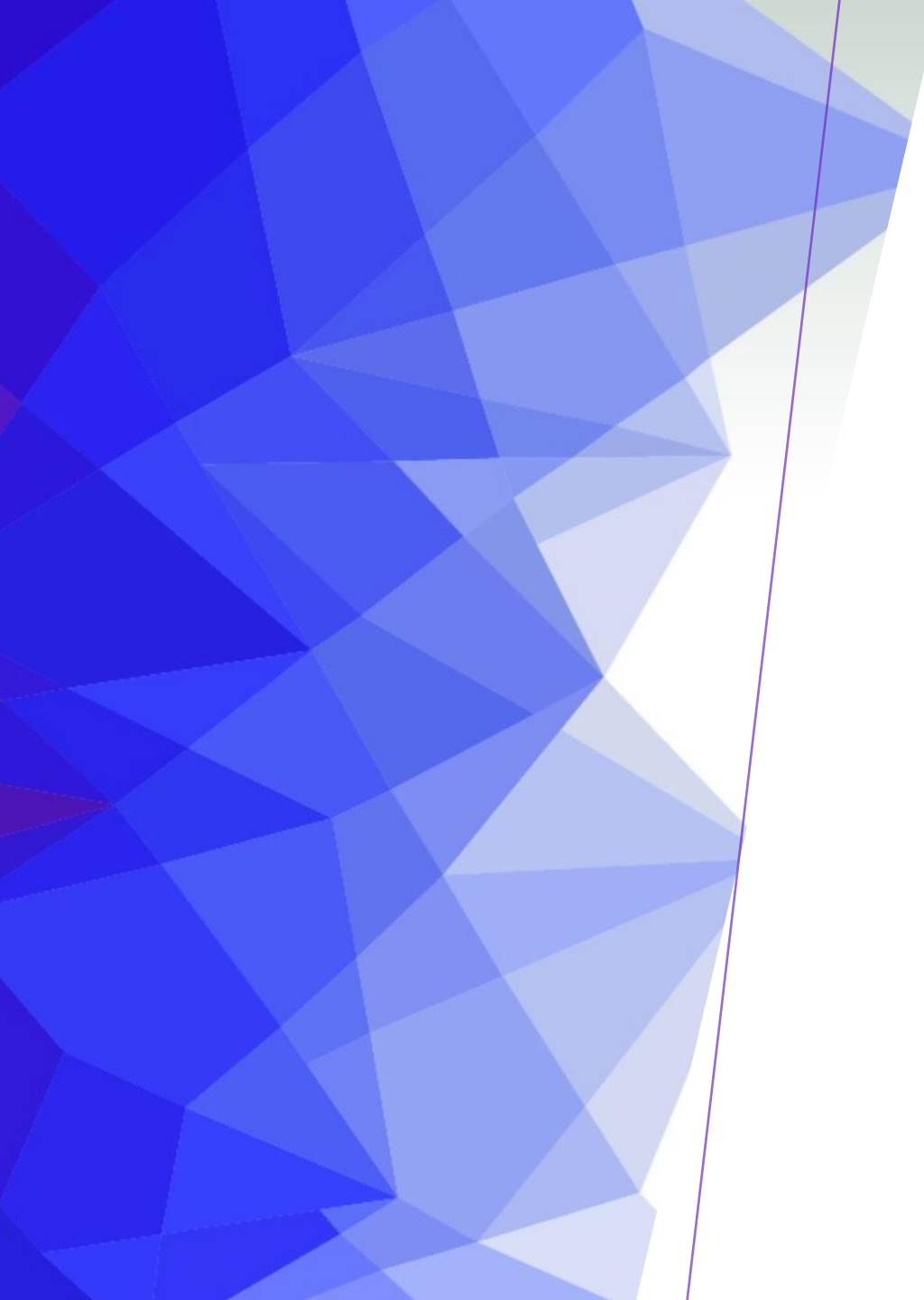
```
</html>
```

TRY IT OUT!

- Save the page as **mypage2.html** and open the file in your Internet browser.
- The page should look like this:



OTHER HTML TAGS

A decorative graphic on the left side of the slide, consisting of a complex, overlapping pattern of blue triangles and polygons in various shades of blue, creating a mosaic-like effect.

OTHER HTML TAGS

- As mentioned before, there are logical styles that describe what the text should be and physical styles which actually provide physical formatting
- It is recommended to use the logical tags and use style sheets to style the text in those tags.

OTHER HTML TAGS

- Logical Tags

Tag	Description
<abbr>	Defines an abbreviation
<acronym>	Defines an acronym
<address>	Defines an address element
<cite>	Defines a <i>citation</i>
<code>	Defines <code>computer code text</code>
<blockquote>	Defines a long quotation
	Defines text
<dfn>	Defines a <i>definition</i> term
	Defines <i>emphasized</i> text
<ins>	Defines inserted text
<kbd>	Defines keyboard text
<pre>	Defines preformatted text
<q>	Defines a short quotation
<samp>	Defines sample computer code
	Defines strong text
<var>	Defines a <i>variable</i>

OTHER HTML TAGS

- Physical Tags

Tag	Description
	Defines bold text
<big>	Defines big text
<i>	Defines <i>italic</i> text
<small>	Defines small text
<sup>	Defines ^{superscripted} text
<sub>	Defines _{subscripted} text
<tt>	Defines teletype text
<u>	Deprecated. Use styles instead

HTML CHARACTER ENTITIES

HTML CHARACTER ENTITIES

- Some characters have a special meaning in HTML, like the less than sign (<) that defines the start of an HTML tag.
- If we want the browser to **actually display** these characters, we must insert **character entities** in place of the **actual characters** themselves.

HTML CHARACTER ENTITIES

- The most common character entities are:

Result	Description	Entity Name	Entity Number
	non-breaking space	 	
<	less than	<	<
>	greater than	>	>
&	ampersand	&	&
"	quotation mark	"	"
'	apostrophe	' (does not work in IE)	'

HTML CHARACTER ENTITIES

- A character entity has three parts:
 - an ampersand (&),
 - an entity name or an entity number, and finally
 - a semicolon (;).
- The **&** means we are beginning a special character, the **;** means ending a special character and the letters in between are sort of an abbreviation for what it's for.

HTML CHARACTER ENTITIES

- To display a less than sign in an HTML document we must write: **<** or **<**;
- The **advantage** of using a **name** instead of a **number** is that a name is **easier to remember**.
- The **disadvantage** is that **not all browsers support the newest entity names**, while the support for entity numbers is very good in almost all browsers.
- **Note:** that entities are case sensitive.

NON-BREAKING SPACE

- The most common character entity in HTML is the non-breaking space ** **;
- Normally HTML will truncate spaces in your text.
- If you add 10 spaces in your text, HTML will remove 9 of them.
- To add spaces to your text, use the ** ** character entity

NON-BREAKING SPACE

- For example:

This Code	Would Display
<code><p> This code would appear as this.</p></code>	This code would appear as this.

This Code	Would Display
<code><p> This code &nbsp;&nbsp;&nbsp;&nbsp; would appear with three extra spaces.</p></code>	This code would appear with three extra spaces.

Some useful character entities are outlined [here](#).

HTML FONTS

HTML FONTS

- The **** tag in HTML is deprecated.
- The W3C has **removed** the tag from its recommendations
- In future versions of HTML, **Cascading Style Sheets**, or **CSS**, are to be used to define the layout and display properties of HTML elements.
- The **** Tag Should **NOT** be used.



HTML BACKGROUNDS

BACKGROUNDS

- The **<body>** tag has two attributes where you can specify backgrounds.
- The background can be a color (note the spelling) or an image.

BACKGROUNDS

- The **bgcolor attribute** specifies a background-color for an HTML page.
- The value of this attribute can be a hexadecimal number, an RGB value, or a color name:

```
<body bgcolor="#000000">
```

```
<body bgcolor="rgb(0,0,0)">
```

```
<body bgcolor="black">
```

- The lines above all set the background-color to black.

BACKGROUND

- The **background attribute** can also specify a **background-image** for an HTML page.
- The **value** of this attribute is the **URL** of the image you want to use.

BACKGROUND

- If the image is smaller than the browser window, the image will repeat itself until it fills the entire browser window.

```
<body background="clouds.gif">
```

```
<body  
background="http://profdevtrain.austincc  
.edu/html/graphics/clouds.gif">
```

- The URL can be relative (as in the first line above) or absolute (as in the second line above).
- You can also refer to a physical address such as
"=D:\myWork\SOD316C\Unit 1\image1.jpg"

(note that the extension of the file is included)



BACKGROUND

- If you want to use a background image, you should ask the following questions:
 - Will the background image increase the loading time too much?
 - Will the background image look good with other images on the page?
 - Will the background image look good with the text colors on the page?
 - Will the background image look good when it is repeated on the page?
 - Will the background image take away the focus from the text?

TRY IT OUT!

```
<html>
<head>
  <title>My First Webpage</title>
</head>
<body
background="http://profdevtrain.austincc.edu/html/graphics/clouds.gif" bgcolor="#EDDD9E">
  <h1 align="center">My First Webpage</h1>
  <p>Welcome to my <strong>first</strong>
webpage. I am writing this page using a text
editor and plain old html.</p>
  <p>By learning html, I'll be able to create
webpages like a <del>beginner</del>
pro....<br>
  which I am of course.</p>
</body>
</html>
```



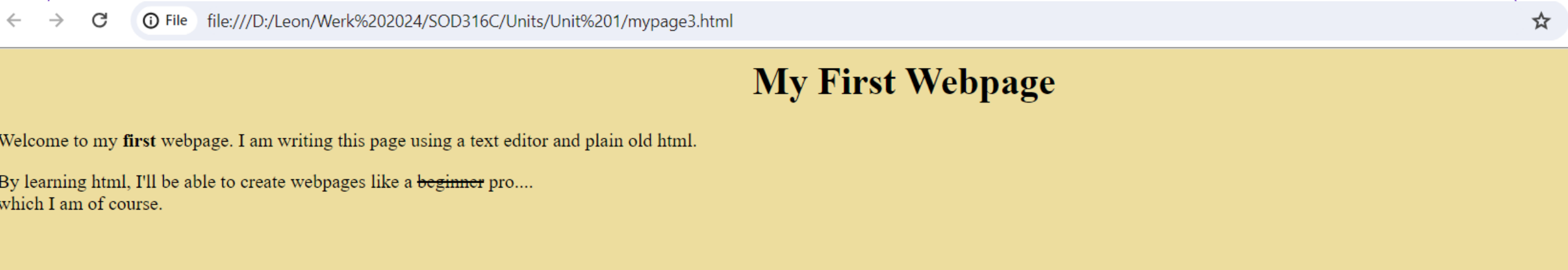
TRY IT OUT!

Save your page as **mypage3.html** and view it in your browser.



TRY IT OUT!

- The page should look like this:



HTML COLOURS



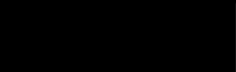
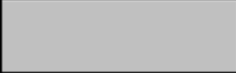
COLOR VALUES

- Colors are defined using a hexadecimal notation for the combination of red, green, and blue (**RGB**) color values
- The lowest value that can be given to one light source is **0 (hex #00)**.
- The highest value is **255 (hex #FF)**.

A red 3D location pin is positioned on a road that curves into the distance under a clear blue sky. The pin is a simple, stylized shape with a circular hole in the center.

COLOUR VALUES

- This table shows the result of combining red, green, and blue:

Color	Color HEX	Color RGB
	#000000	rgb(0,0,0)
	#FF0000	rgb(255,0,0)
	#00FF00	rgb(0,255,0)
	#0000FF	rgb(0,0,255)
	#FFFF00	rgb(255,255,0)
	#00FFFF	rgb(0,255,255)
	#FF00FF	rgb(255,0,255)
	#C0C0C0	rgb(192,192,192)
	#FFFFFF	rgb(255,255,255)



COLOUR VALUES

- A collection of color names is supported by most browsers.
- To view a table of color names that are supported by most browsers visit [this web page](#).

Color	Color HEX	Color Name
	#F0F8FF	AliceBlue
	#FAEBD7	AntiqueWhite
	#7FFFD4	Aquamarine
	#000000	Black
	#0000FF	Blue
	#8A2BE2	BlueViolet
	#A52A2A	Brown

A red 3D location pin is positioned on a dark road with white dashed lines, set against a light blue sky background. The pin is a simple teardrop shape with a circular hole in the center.

COLOUR VALUES

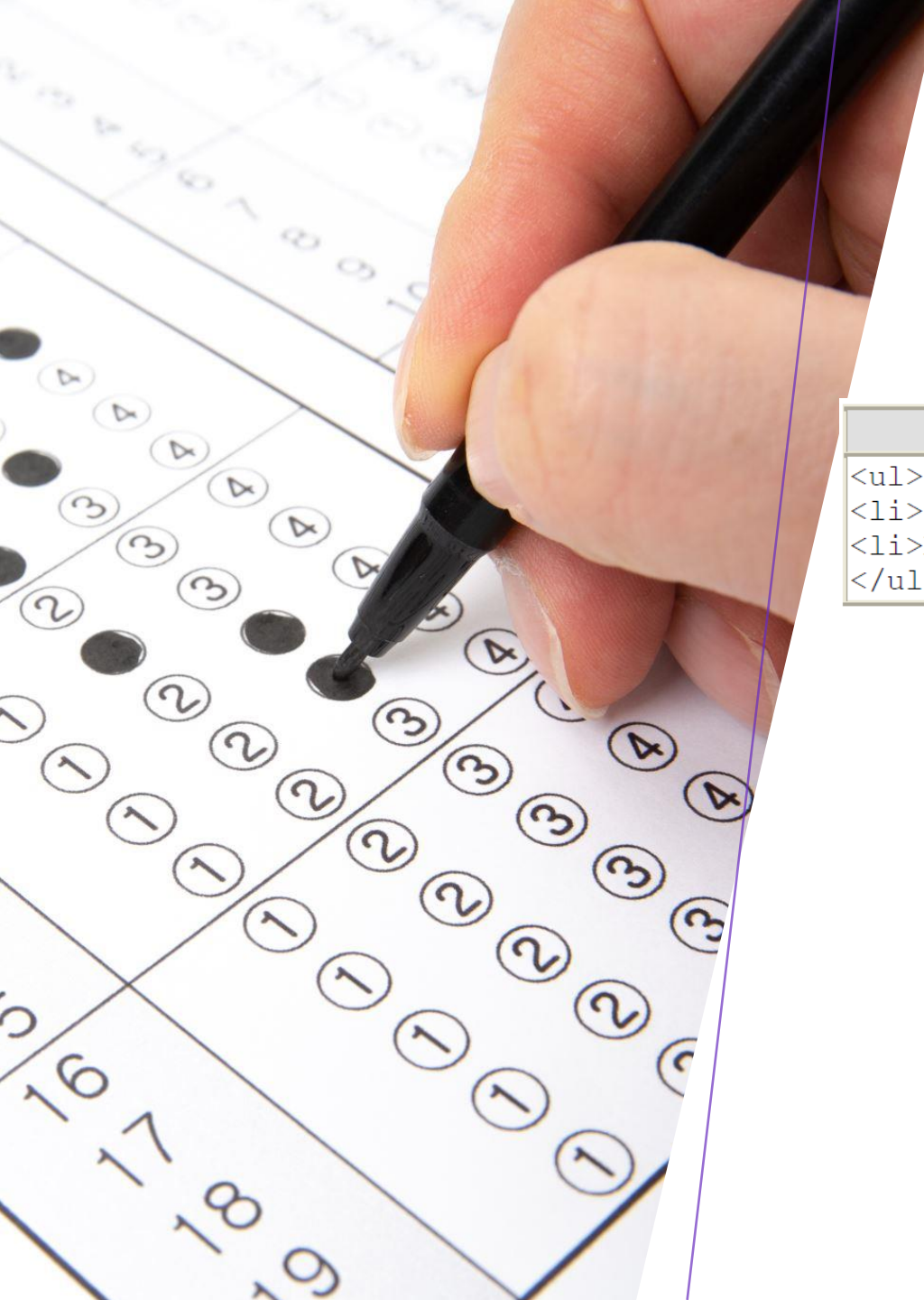
- The combination of Red, Green and Blue values from 0 to 255 gives a total of more than **16 million different colors** to play with ($256 \times 256 \times 256$).
- Most modern monitors can display at least 16,384 different colors.

HTML LISTS



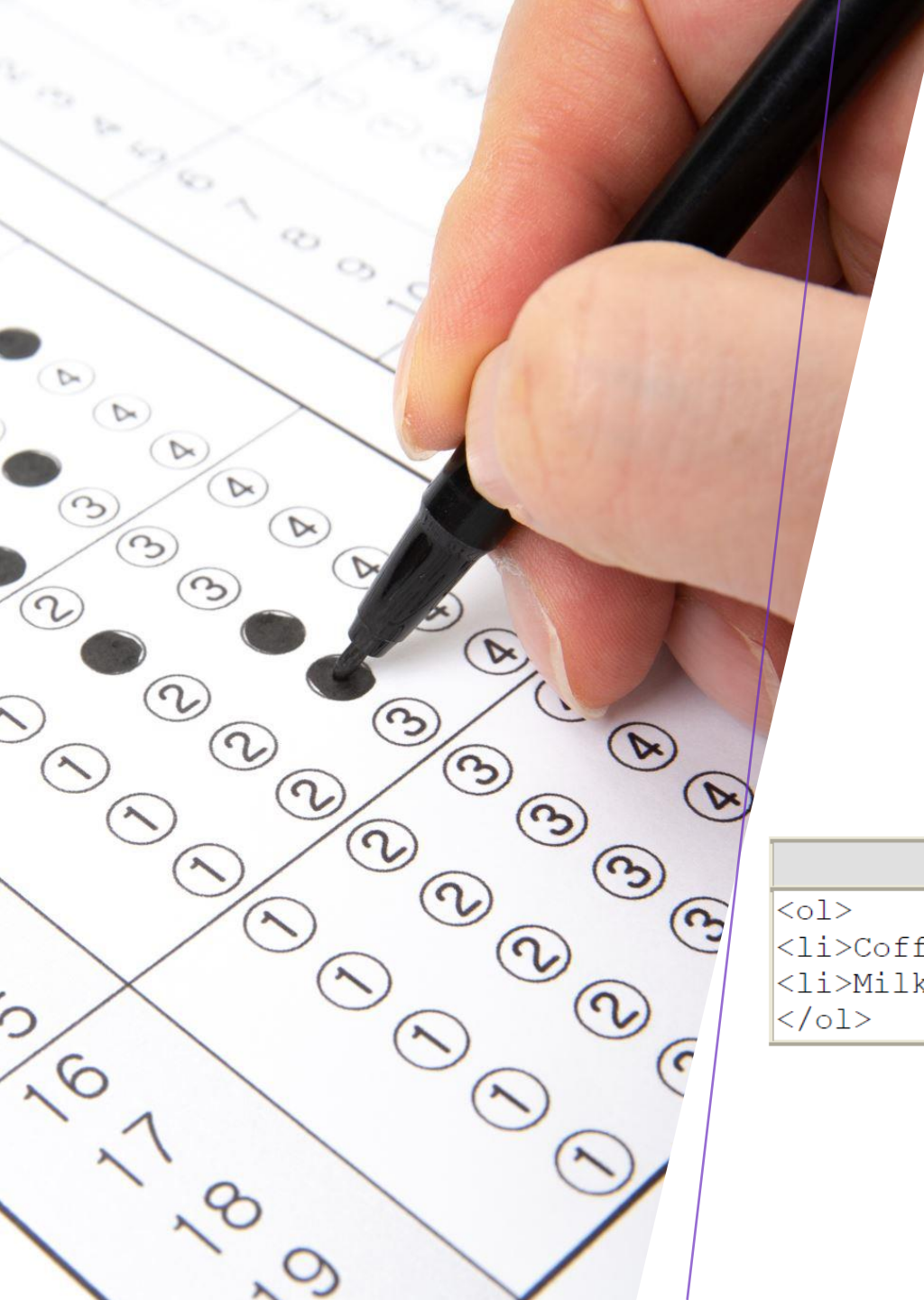
UNORDERED LISTS

- HTML provides a simple way to show **unordered lists (bullet lists)** or **ordered lists (numbered lists)**.
- An **unordered list** is a list of items marked with **bullets** (typically small black circles).
- An unordered list starts with the **** tag.
- Each list item starts with the **** tag.



UNORDERED LISTS

This Code	Would Display
<pre> Coffee Milk </pre>	▪ Coffee ▪ Milk



ORDERED LISTS

- An **ordered list** is also a list of items.
- The **list items** are marked with **numbers**.
- An ordered list starts with the **** tag.
- Each list item starts with the **** tag.

This Code	Would Display
<pre> Coffee Milk </pre>	<pre>1. Coffee 2. Milk</pre>

DEFINITION LISTS

- Definition lists consist of two parts: a term and a description.
- To mark up a definition list, you need three HTML elements; a container `<dl>`, a definition term `<dt>`, and a definition description `<dd>`.

This Code	Would Display
<pre><dl> <dt>Cascading Style Sheets</dt> <dd>Style sheets are used to provide presentational suggestions for documents marked up in HTML. </dd> </dl></pre>	Cascading Style Sheets Style sheets are used to provide presentational suggestions for documents marked up in HTML.

TRY IT OUT!

```
<html>
```

```
... head and title section here ...
```

```
<body bgcolor="#EDDD9E">
```

```
  <h1 align="center">My First Webpage</h1>
```

```
  <p>Welcome to my <strong>first</strong> webpage. I am writing  
this page using a text editor and plain old html.</p>
```

```
  <p>By learning html, I'll be able to create web pages like a  
pro....<br>
```

```
  which I am of course.</p>
```

```
  Here's what I've learned:
```

```
  <ul>
```

```
    <li>How to use HTML tags</li>
```

```
    <li>How to use HTML colors</li>
```

```
    <li>How to create Lists</li>
```

```
  </ul>
```

```
  </body>
```

```
</html>
```

TRY IT OUT!

- Save your page as `mypage4.html` and view it in your browser.



TRY IT OUT!

- Your page should look like this:



HTML LINKS

HTML LINKS

- HTML uses the **<a>** anchor tag to create a link to another document or web page.
- An **anchor** can point to any resource on the Web: an HTML page, an image, a sound file, a movie, etc.
- The syntax of creating an anchor:

Text to be displayed

HTML LINKS

- The **<a>** tag is used to create an anchor to link from, the **href** attribute is used to tell the address of the document or page we are linking to.
- The words between the open and close of the anchor tag will be displayed as a hyperlink.

This Code	Would Display
<code>Visit ACC!</code>	<u>Visit ACC!</u>



THE ANCHOR TAG AND THE HREF ATTRIBUTE

- An **anchor** can **point** to **any resource** on the **Web**: an HTML page, an image, a sound file, a movie, etc
- The syntax of creating an anchor:

`<a href="url">Text to be displayed</a>`

- The **`<a>`** tag is used to create an anchor to link from, the href attribute is used to tell the address of the document or page we are linking to, and the words between the open and close of the anchor tag will be displayed as a hyperlink

THE TARGET ATTRIBUTE

- With the target attribute, you can define where the linked document will be opened.
- By default, the link will open in the current window.
- The code below will open the document in a new browser window:

```
<a href=http://www.CUT.ac.za/  
target="_blank">Visit CUT!</a>
```



EMAIL LINKS

- To create an email link, you will use mailto: plus your email address.
- Here is an example of a link to the CUT Help Desk:

```
<a href="mailto:helpdesk@cut.ac.za">Email  
Help Desk</a>
```

- To add a subject for the email message, you would add **?subject=** after the email address. For example:

```
<a href="mailto:helpdesk@austincc.edu  
?subject=Email Assistance">Email Help  
Desk</a>
```

HTML IMAGES

THE IMAGE TAG AND THE SRC ATTRIBUTE

- The `<img>` tag is empty, which means that it contains attributes only and it has no closing tag.
- To display an image on a page, you need to use the **src** attribute.
- **src** stands for "source".
- The value of the **src** attribute is the URL of the image you want to display on your page.

THE IMAGE TAG AND THE SRC ATTRIBUTE

- The syntax of defining an image looks like this in its most basic form:

This Code	Would Display
<code></code>	

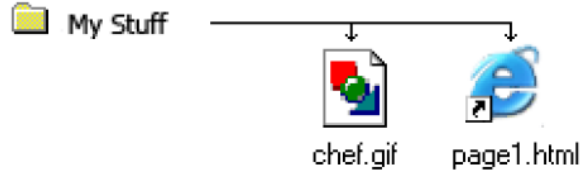
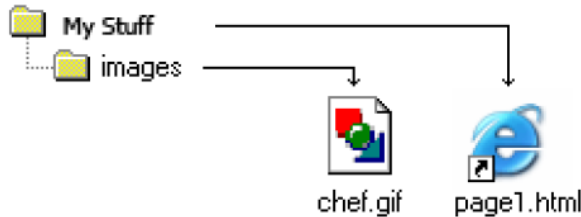
THE IMAGE TAG AND THE SRC ATTRIBUTE

- Not only does the source attribute specify what image to use, but where the image is located.
- The image used here, graphics/chef.gif, means that the browser will look for the image name chef.gif in a graphics folder in the same folder as the html document itself.

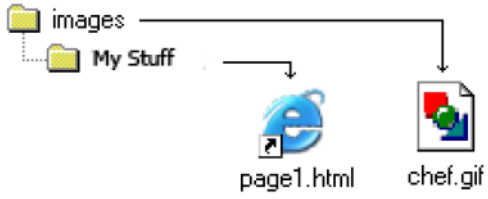
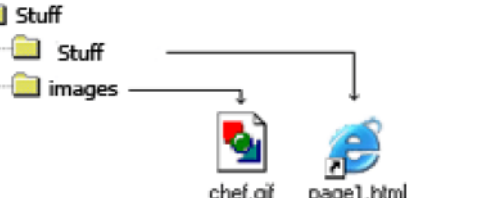
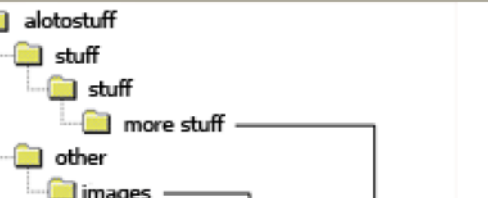
This Code	Would Display
<pre></pre>	

THE IMAGE TAG AND THE SRC ATTRIBUTE

- The browser puts the image where the image tag occurs in the document.
- If you put an image tag between two paragraphs, the browser shows the first paragraph, then the image, and then the second paragraph.

	<code>src="chef.gif"</code> means that the image is in the same folder as the html document calling for it.
	<code>src="images/chef.gif"</code> means that the image is one folder down from the html document that called for it. This can go on down as many layers as necessary.

THE IMAGE TAG AND THE SRC ATTRIBUTE

	<code>src="../../../chef.gif"</code> means that the image is in one folder up from the html document that called for it.
	<code>src="../../chef.gif"</code> means that the image is two folders up from the html document that called for it.
	<code>src="../images/chef.gif"</code> means that the image is one folder up and then another folder down in the images directory.
	<code>src="../../../other/images/chef.gif"</code> means this goes multiple layers up.

THE ALT ATTRIBUTE

- The alt attribute is used to define an alternate text for an image.
- The value of the alt attribute is author-defined text:

```

```

THE ALT ATTRIBUTE

- The alt attribute tells the reader what he or she is missing on a page if the browser can't load images.
- The browser will then display the alternate text instead of the image.
- It is a good practice to include the alt attribute for each image on a page, to improve the display and usefulness of your document for people who have text-only browsers or use screen readers.

IMAGE DIMENSIONS

- When you have an image, the browser usually figures out how big the image is all by itself.
- If you put in the image dimensions in pixels however, the browser simply reserves a space for the image, then loads the rest of the page.
- Once the entire page is loads it can go back and fill in the images.

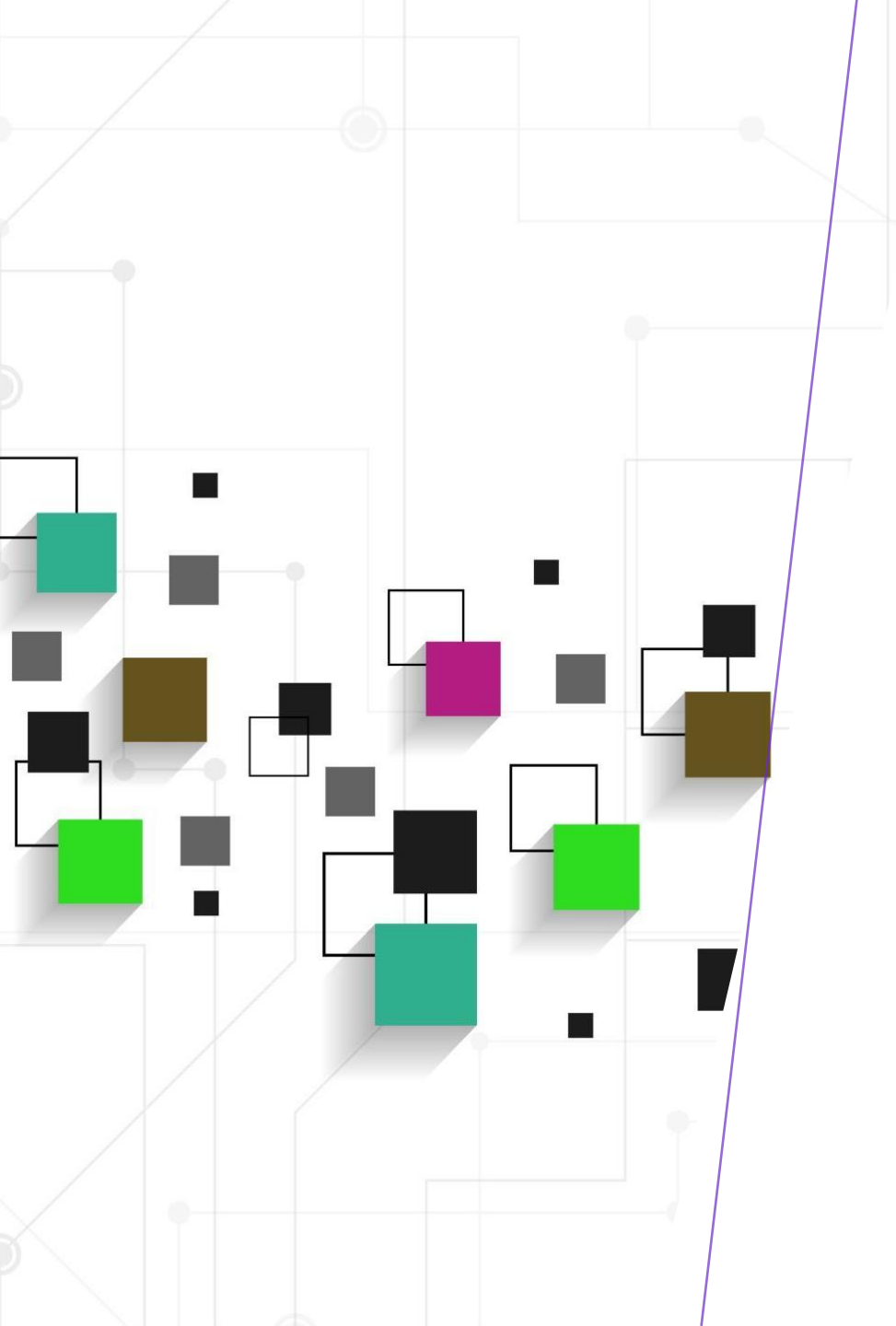
IMAGE DIMENSIONS

- Without dimensions, when it runs into an image, the browser has to pause loading the page, load the image, then continue loading the page.
- The chef image would then be:

```

```

HTML TABLES



TABLES

- Tables are defined with the **<table>** tag.
- A table is divided into rows (with the **<tr>** tag), and each row is divided into data cells (with the **<td>** tag).
- The letters **td** stands for table data, which is the content of a data cell.
- A data cell can contain text, images, lists, paragraphs, forms, horizontal rules, tables, etc.

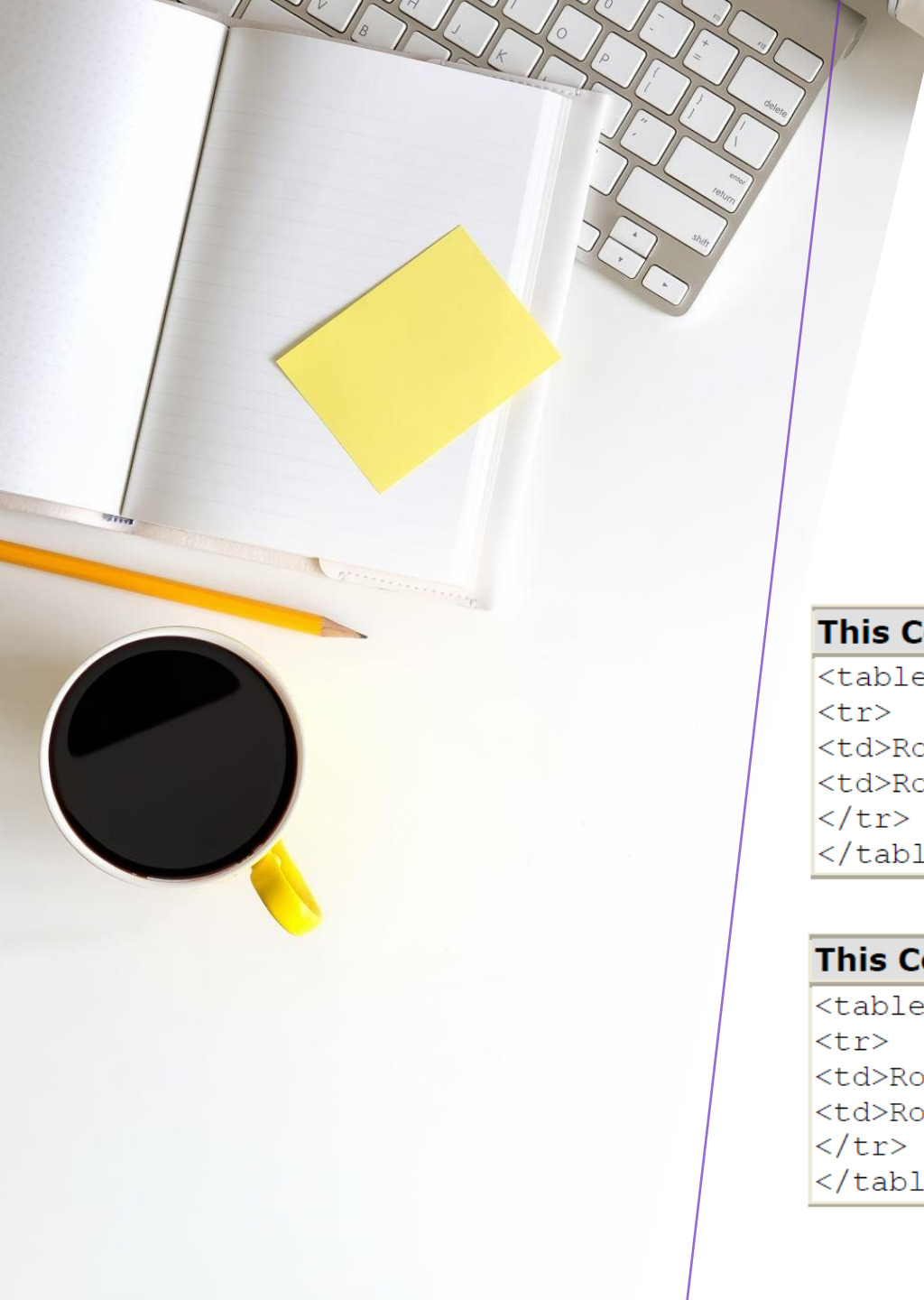
TABLES

This Code

```
<table>
<tr>
<td>row 1, cell 1</td>
<td>row 1, cell 2</td>
</tr>
<tr>
<td>row 2, cell 1</td>
<td>row 2, cell 2</td>
</tr>
</table>
```

Would Display

row 1, cell 1 row 1, cell 2
row 2, cell 1 row 2, cell 2



TABLES AND THE BORDER ATTRIBUTE

- To display a table with borders, you will use the **border** attribute.

This Code	Would Display		
<pre><table border="1"> <tr> <td>Row 1, cell 1</td> <td>Row 1, cell 2</td> </tr> </table></pre>	<table><tr><td>row 1, cell 1</td><td>row 1, cell 2</td></tr></table>	row 1, cell 1	row 1, cell 2
row 1, cell 1	row 1, cell 2		

This Code	Would Display		
<pre><table border="5"> <tr> <td>Row 1, cell 1</td> <td>Row 1, cell 2</td> </tr> </table></pre>	<table><tr><td>row 1, cell 1</td><td>row 1, cell 2</td></tr></table>	row 1, cell 1	row 1, cell 2
row 1, cell 1	row 1, cell 2		



CELL PADDING AND SPACING

- The `<table>` tag has two attributes known as cell spacing and cellpadding.
- Here is a table example without these properties (These properties may be used separately or together):

This Code	Would Display				
<pre><table border="1"> <tr> <td>some text</td> <td>some text</td> </tr> <tr> <td>some text</td> <td>some text</td> </tr> </table></pre>	<table><tr><td>some text</td><td>some text</td></tr><tr><td>some text</td><td>some text</td></tr></table>	some text	some text	some text	some text
some text	some text				
some text	some text				



CELL PADDING AND SPACING

- **Cell spacing** is the pixel width between the individual data cells in the table (The thickness of the lines making the table grid).
- The default is zero. If the border is set at 0, the cell spacing lines will be invisible.

This Code

```
<table border="1" cellspacing="5">
<tr>
<td>some text</td>
<td>some text</td>
</tr><tr>
<td>some text</td>
<td>some text</td>
</tr>
</table>
```

Would Display

some text	some text
some text	some text



CELL PADDING AND SPACING

- **Cellpadding** is the pixel space between the cell contents and the cell border.
- The default for this property is also zero.
- This feature is not used often, but sometimes comes in handy when you have your borders turned on and you want the contents to be away from the border a bit for easy viewing.
- Cellpadding is invisible, even with the border property turned on. Cellpadding can be handled in a style sheet.



CELL PADDING AND SPACING

This Code	Would Display				
<pre><table border="1" cellpadding="10"> <tr> <td>some text</td> <td>some text</td> </tr><tr> <td>some text</td> <td>some text</td> </tr> </table></pre>	<table><tr><td>some text</td><td>some text</td></tr><tr><td>some text</td><td>some text</td></tr></table>	some text	some text	some text	some text
some text	some text				
some text	some text				



CELL PADDING AND SPACING

- Useful table tags:

Tag	Description
<table>	Defines a table
<th>	Defines a table header
<tr>	Defines a table row
<td>	Defines a table cell
<caption>	Defines a table caption
<colgroup>	Defines groups of table columns
<col>	Defines the attribute values for one or more columns in a table



CELL PADDING AND SPACING

- The width attribute can be used to define the width of your table.
- It can be defined as a fixed width or a relative width.
- A fixed table width is one where the width of the table is specified in pixels.
- For example, this code, **<table width="550">**, will produce a table that is 550 pixels wide.



CELL PADDING AND SPACING

- A relative table width is specified as a percentage of the width of the visitor's viewing window.
- This code, `<table width="80%">`, will produce a table that occupies 80 percent of the screen.
- Which table size attribute do you think is better to use and why?



PUTTING IT ALL TOGETHER

- Let's create a simple page by putting together everything you learned in this lesson.
- Refer to the document "Unit 1 – Putting it all together".